

OAC Catalog ACL Extraction Pipeline

Automated permissions intelligence — Python notebooks, Apache Spark, Oracle AIDP, OAC, and ADW

198

ACL Rows

5

Catalog Types

10

Enrichments

6

Report Views

THE CHALLENGE

Oracle deprecated the Catalog Manager desktop tool with REST API as the only replacement.

The only supported path to bulk-audit OAC permissions was eliminated. For enterprise environments managing workbooks, datasets, dataflows, folders, and connections across hundreds of users, this left a real governance gap. The OAC REST API was the only remaining path — but required custom engineering to become a reliable, scheduled, and reportable solution.

WHAT WE BUILT — MEDALLION PIPELINE – via Oracle AI Data Platform



TECHNICAL HIGHLIGHTS

Python Notebooks · AIDP Workbench
Three-notebook Medallion pipeline on a Spark cluster in the AIDP workspace. Bronze extracts raw ACLs, Silver applies 10 PySpark enrichments, Gold unions four aggregated views into a single ADW table.

OAC REST API · ACL Extraction
Iterates all five catalog object types with full pagination. Nested JSON responses flattened to one row per subject-object-permission. Proactive 5-min token refresh prevents auth failures mid-run.

ADW Write Pattern · External Catalog
saveAsTable() with 3-part path, mode=overwrite, overwriteSchema=True. No Delta format (ADW unsupported) and no schema DDL (502 from AIDP Metastore). Solved once in Bronze, reused across Silver and Gold.

Silver Enrichments · Risk Scoring
PERMISSION_LEVEL normalizes raw codes to Read / Write / Full Control. RISK_SCORE (0-10) weights permission level, account category, and catalog type. RISK_LEVEL buckets to Low / Medium / High / Critical.

Custom DV Canvas Authoring
Canvases 5 and 6 built as raw DV XML and JSON fragments — not drag-and-drop — enabling RISK_LEVEL conditional formatting, per-tier color encoding, and layout precision beyond the OAC low-code builder.

OAC Token Auth · tokens.json
Uses OAC's built-in profile token download. Token via OAC Profile → Access Tokens → tokens.json. Manual daily refresh required; OCI Vault integration in progress to eliminate all manual token handling.

THE REPORT — SIX ADMIN VIEWS

01 Summary Overview
High-level access counts across all catalog types and subjects. Entry point for any admin review.

02 Access Matrix
Pivot of users vs. object types. Shows breadth of access at a glance across all five catalog types.

03 High Privilege Alert
Every Full Control and elevated permission surfaced automatically — driven by PERMISSION_LEVEL from Silver.

04 By Owner
Object coverage per owner. Flags orphaned assets and accounts with unusual ownership concentration.

05 Permission Analytics
Multi-dimension breakdown by level, catalog type, account category, and owner. Pattern recognition at scale.

06 Risk Dashboard
RISK_SCORE and RISK_LEVEL driven visuals, built from raw DV XML. Turns 198 rows into a prioritized action list.

WHAT'S NEXT

In Progress
Token Automation
CLIENT_SECRET from OCI Vault enables a fully automated IDCS client credentials flow for unattended nightly runs.

Planned
Workflow Scheduling
Bronze → Silver → Gold wired as a 3-task AIDP Workflow with dependency sequencing and a nightly schedule.

Planned
Report Dataset Refresh
Repoint OAC canvases from Bronze to Silver and Gold to surface all enriched columns across the six views.

Planned
Repo Tag v1.0
Commit tested Gold .ipynb notebook and tag aidp-notebooks at v1.0 once all layers run on schedule.